

Bibliography

A selection of publications and websites which contain useful information on pharmaceutical excipients is listed below:

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Abbreviations

Some units, terms, and symbols are not included in this list as they are defined in the text. Common abbreviations have been omitted. The titles of journals are abbreviated according to the general style of the *Index Medicus*.

≈	approximately.	IV	intravenous.
Ad	Addendum.	J	joule(s).
ADI	acceptable daily intake.	JP	Japanese Pharmacopeia.
approx	approximately.	JPE	Japanese Pharmaceutical Excipients
atm	atmosphere.	kcal	kilocalorie(s).
BAN	British Approved Name.	kg	kilogram(s).
bp	boiling point.	kJ	kilojoule(s).
BP	British Pharmacopoeia.	kPa	kilopascal(s).
BS	British Standard (specification).	L	liter(s).
BSI	British Standards Institution.	LAL	<i>Limulus</i> amoebocyte lysate.
cal	calorie(s).	LC ₅₀	a concentration in air lethal to 50% of the specified animals on inhalation.
CAS	Chemical Abstract Service.	LD ₅₀	a dose lethal to 50% of the specified animals or microorganisms.
CFC	chlorofluorocarbon.	Ld _{Lo}	lowest lethal dose for the specified animals or microorganisms.
cfu	colony-forming unit	m	meter(s).
cm	centimeter(s).	m ²	square meter(s).
cm ²	square centimeter(s).	m ³	cubic meter(s).
cm ³	cubic centimeter(s).	M	molar.
cmc	critical micelle concentration.	max	maximum.
CNS	central nervous system.	MCA	Medicines Control Agency (UK).
cP	centipoise(s).	mg	milligram(s).
cSt	centistoke(s).	MIC	minimum inhibitory concentration.
CTFA	Cosmetic, Toiletry, and Fragrance Association.	min	minute(s) <i>or</i> minimum.
d	particle diameter (d ₁₀ at 10 percentile; d ₅₀ at 50 percentile; d ₉₀ at 90 percentile).	mL	milliliter(s).
D&C	designation applied in USA to dyes permitted for use in drugs and cosmetics.	mm	millimeter(s).
DoH	Department of Health (UK).	mM	millimolar.
DSC	differential scanning calorimetry.	mm ²	square millimeter(s).
EC	European Community.	mm ³	cubic millimeter(s).
e.g.	<i>exempli gratia</i> , 'for example'.	mmHg	millimeter(s) of mercury.
EINECS	European Inventory of Existing Commercial Chemical Substances.	mmol	millimole(s).
<i>et al</i>	<i>et alii</i> , 'and others'.	mN	millinewton(s).
EU	European Union.	mol	mole(s).
FAO	Food and Agriculture Organization of the United Nations.	mp	melting point.
FAO/WHO	Food and Agriculture Organization of the United Nations <i>and</i> the World Health Organization.	mPa	millipascal(s).
FCC	Food Chemicals Codex.	MPa	megapascal(s).
FDA	Food and Drug Administration of the USA.	μg	microgram(s).
FD&C	designation applied in USA to dyes permitted for use in foods, drugs, and cosmetics.	μm	micrometer(s).
FFBE	Flat face beveled edge.	N	newton(s) <i>or</i> normal (concentration).
g	gram(s).	NIR	near-infrared.
GMP	Good Manufacturing Practice.	nm	nanometer(s).
GRAS	generally recognized as safe by the Food and Drug Administration of the USA.	o/w	oil-in-water.
HC	hydrocarbon.	o/w/o	oil-in-water-in-oil.
HCFC	hydrochlorofluorocarbon.	Pa	pascal(s).
HFC	hydrofluorocarbon.	pH	the negative logarithm of the hydrogen ion concentration.
HIV	human immunodeficiency virus.	PhEur	European Pharmacopeia.
HLB	hydrophilic-lipophilic balance.	pK _a	the negative logarithm of the dissociation constant.
HSE	Health and Safety Executive (UK).	pph	parts per hundred.
i.e.	<i>id est</i> , 'that is'.	ppm	parts per million.
IM	intramuscular.	psia	pounds per square inch absolute.
INN	International Nonproprietary Name.	R	reflectance.
IP	intraperitoneal.	RDA	recommended dietary allowance (USA).
ISO	International Organization for Standardization.	rpm	revolutions per minute.
IU	International Units.	s	second(s).
		SC	subcutaneous.
		SEM	scanning electron microscopy <i>or</i> scanning electron microphotography.